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# First in the World Complex Through Tubing Sand Control System (Screen Expansion) Installation Via Combination of E-Line Conveyance with Tractor and Stroker

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## Abstract

Well K was completed in October 2016 as a gas producer, with a 3-1/2" cemented monobore completion, an 80-degree deviation, and 1,300 m of tangent section. As a monobore completion, the well was completed with commingled production from 5 different reservoir layers. The well produced steadily at 6 MMSCFD since completion but was choked down to 2 MMSCFD in October 2022 due to high sand production. Further evaluation (a sand detection log) was conducted to determine the source of the sand-producing zone. Over time, the well was shut in due to excessive sand production. Given the high well deviation and long tangent section, selecting an appropriate sand control option is never a straightforward process. The main selection criteria shall consider the longevity of well productivity without requiring frequent well interventions to replace the screen or the capability to deploy a complete one-way through-tubing system in a challenging environment (i.e. TTGP). This is because any intervention in the well would necessitates the use of Coiled Tubing or Electric Line with tractoring which involve high cost and operational risks.

After a comprehensive evaluation, screen expansion sand control system was chosen over other alternatives due to its proven capability to deliver long-term sand control performance downhole. Considering the high well deviation and limited deck space on the platform, an electric line-conveyed system with a tractor and well stroker was preferred over Coiled Tubing approach, offering a minimal footprint and removing the need for vessel support. Detailed planning, starting with an on-paper review, followed by a System Integration Test and pre-mobilization inspection and testing was carried out to address potential operational issues prior to mobilization. However, the absence of global experience in conveying long through-tubing expandable screen assemblies across extended perforation intervals using electric line conveyance resulted in certain operational risks not being fully captured during the design phase. As a result, various modifications and enhancements to the operational steps were implemented during execution to ensure continuity until the completion of the operation.

The job was successfully executed with 41-meters of through tubing expanded screen installed across a 35-meter perforation interval. The well was brought back to its initial Technical Potential of 6 MMSCFD

without any sand production post-operation. Numerous lessons learned were captured throughout the operation and there is a list of improvements to be practiced in future similar operations.

This paper will describe the overall job design from the planning phase until the job execution phase, covering specific and novel procedure development associated with efforts to ensure a successful operation, which is recognized as the first similar operation implemented in the world.

## Well and Field Overview

Well K, located off the coast of Sarawak, was completed in October 2016 as a 3-1/2" monobore gas producer with a cemented production string utilizing 9.2 ppf tubing. The completion design provides hydraulic access to five distinct gas-bearing zones and enables commingled production through an integrated tubing-conveyed perforation strategy.

The platform wellhead service area limited to only 117m² of space so there was a lot of limitation associated to bringing big fleet equipment such as Coiled Tubing to the platform where it requires vessel support to place various equipment before able to continue with well access work. In view with limited platform deck space, slickline and electric line equipment are the most common well intervention conveyance to execute various well intervention related operation starting from routine data acquisition operation until complex work such as debris cleanout, perforation, etc.

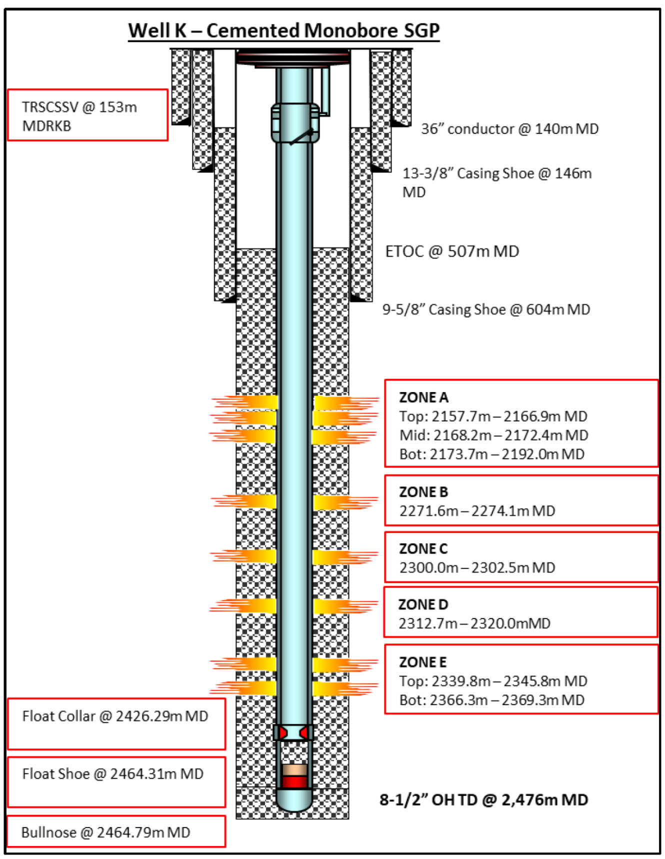



**Figure 1—Current K well sketch. Well accessing to 5 reservoirs in commingle.**

Maximum well deviation is 80.29 deg @ 1356.98m MD with total 1300m of tangent section as shown in Fig. 2 below.

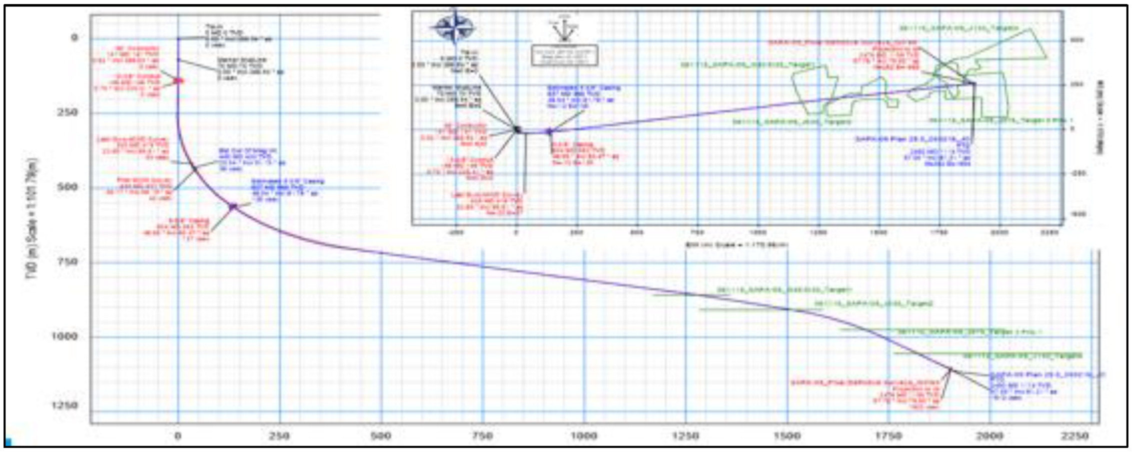


**Figure 2—2D image of Well K showing high well deviation section from 810m MD until 2125m MD.**

According to the well intervention history, there is no record of slickline operations carried out in Well K. This is attributed to the well's high deviation and extended tangent section. Simulations were conducted to evaluate the feasibility of slickline conveyance, with various tool string configurations tested across friction coefficient values ranging from 0.2 to 0.4. However, none of the configurations allowed the slickline tools to reach the bottom of the well at 2,426 m MD. Based on the simulation results, a continuous tractoring force would be required starting at 903 m MD to convey the tools deeper into the wellbore.

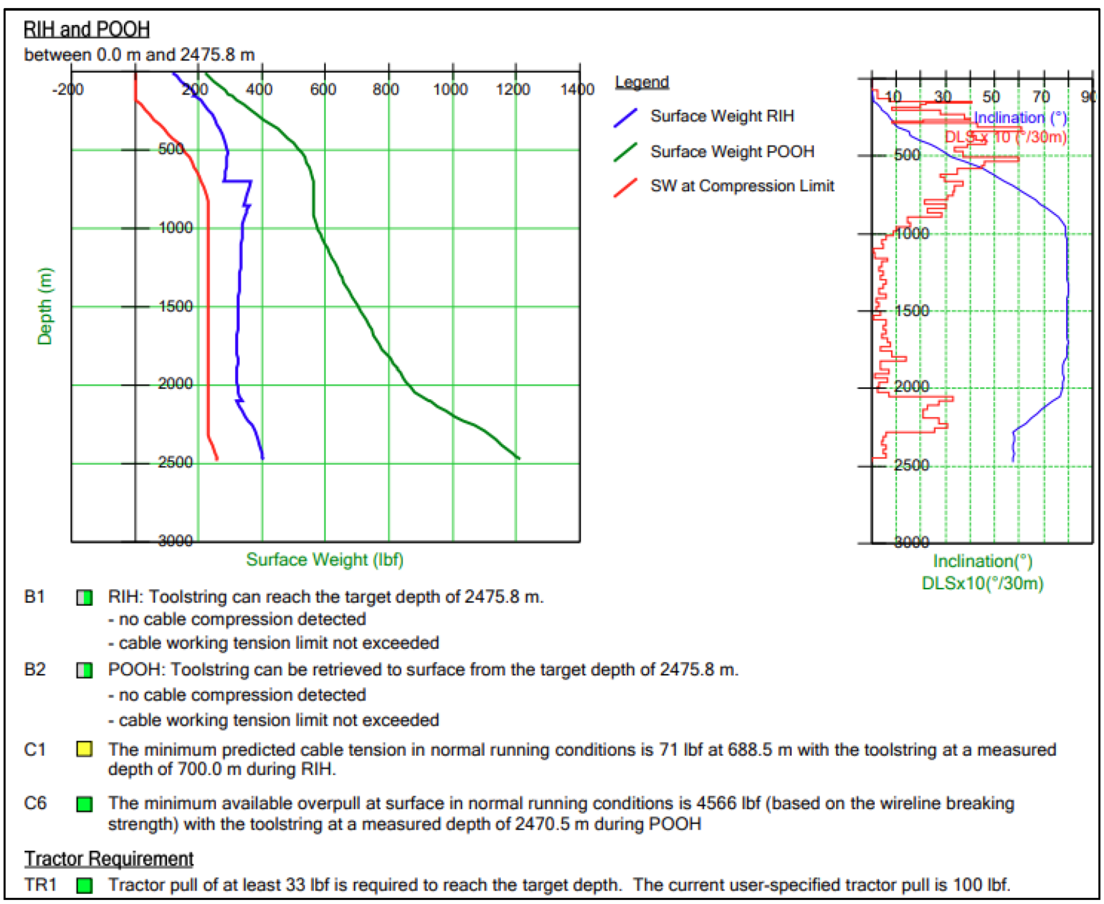


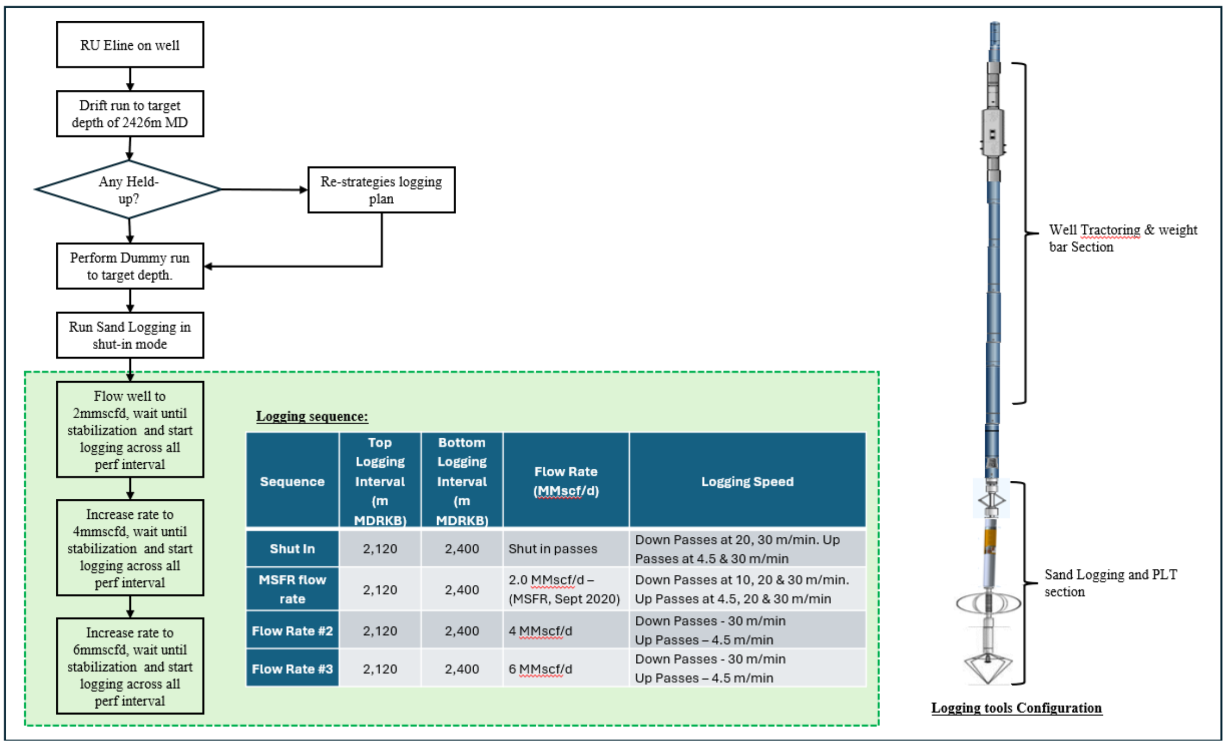

**Figure 3—Well Accessibility Simulation confirm Tractor force is required from 700m MD to convey the tools to the depth.**

## Problem Statement & Challenges

Well K initially produced at a stable rate of 6 MMscfd following completion in October 2016. However, the onset of unexpected sand production led to an unplanned shutdown of the well. This allowed time to determine an optimal flow rate below the Maximum Sand-Free Rate (MSFR). Once the appropriate production parameters were identified, the well was choked back to a rate of 2 MMscfd to limit sand carryover to surface facilities and mitigate potential risks to equipment and operations.

### Produces Sand Detection and Analysis

Due to excessive sand production, a logging operation was conducted to identify the source of sand influx. Given the high deviation of the well, electric-line conveyance with tractor assistance was utilized to deploy Sand Detection Logging Tools across the perforated interval. The objective was to evaluate sand production in relation to flow rates. A step-by-step procedure was established to ensure accurate determination of sand contribution depths.




| Sequence | Top Logging Interval (m MDRKB) | Bottom Logging Interval (m MDRKB) | Flow Rate (MMscf/d) | Logging Speed |
|---|---|---|---|---|
| Shut In | 2,120 | 2,400 | Shut in passes | Down Passes at 20, 30 m/min. Up Passes at 4.5 & 30 m/min |
| MSFR flow rate | 2,120 | 2,400 | 2.0 MMscf/d – (MSFR, Sept 2020) | Down Passes at 10, 20 & 30 m/min. Up Passes at 4.5, 20 & 30 m/min |
| Flow Rate #2 | 2,120 | 2,400 | 4 MMscf/d | Down Passes - 30 m/min Up Passes – 4.5 m/min |
| Flow Rate #3 | 2,120 | 2,400 | 6 MMscf/d | Down Passes - 30 m/min Up Passes – 4.5 m/min |

**Figure 4—Well Sand logging program to confirm the source of sand production.**

After the successful logging operation, a detailed review of the results was carried out. The findings indicated that sand production was predominantly contributed by the upper section of Zone A, while minor sand ingress was detected from the lower section of the same zone, as presented in the following table.

Based on Table 1, there is a clear indication that sand production occurs when the gas flow rate increases to 4 MMscfd. The Maximum Sand-Free Rate (MSFR) was identified at 2 MMscfd, where no sand production was observed. Although logging tools were conveyed through Zones B, C, and D under various flow conditions, these zones were excluded from the results summary as they did not contribute to production, and no sand was detected from them.

In view with the above result, it was concluded that Zone A (Top) are the main contributor to the sand production with minor contribution by Zone A (Bottom).

Table 1—Summary of PLT and Sand Logging Operation Result.

| Zones | Sand Depth / Interval | Gas Production | | | Possible Sand Entry | | |
|---|---|---|---|---|---|---|---|
| | | 2 mmscfd | 4 mmscfd | 6 mmscfd | 2 mmscfd | 4 mmscfd | 6 mmscfd |
| A Top | 2157.7m – 2166.9m MD | 7% | 8% | 11% | No evident. | No evident. | Sand entry detected (fine sand particles) |
| A Middle | 2168.2m – 2172.4m MD | 5% | 5% | 7% | No evident of sand production. | | |
| A Bottom | 2173.7m – 2192.0m MD | 53% | 58% | 62% | No evident. | Possible sand entry. | Minor sand count detected. |
| E | 2339.8m – 2345.8m MD 2366.3m – 2369.3m MD | 31% | 26% | 20% | No evident of sand production. | | |

## Zone A Sand Production Isolation Challenges

Following the identification of sand-producing zones, additional challenges were recognized in implementing zonal isolation. As shown in Figure 6, the spacing between individual perforation intervals within Zone A is limited to a maximum of 1.3 m. Due to this close proximity, executing targeted sand production isolation operations presents significant limitations. This constraint is especially critical when evaluating the feasibility of specific sand control techniques.

a. Specific zone chemical treatment to consolidate the sand. No mechanical isolation tools are available to cater for 1.3m of gap.
b. Tight space between perforation interval limiting Mechanical isolation packer placement in between the perforation interval.
c. Challenges in depth correlation due to tight space between perforation.

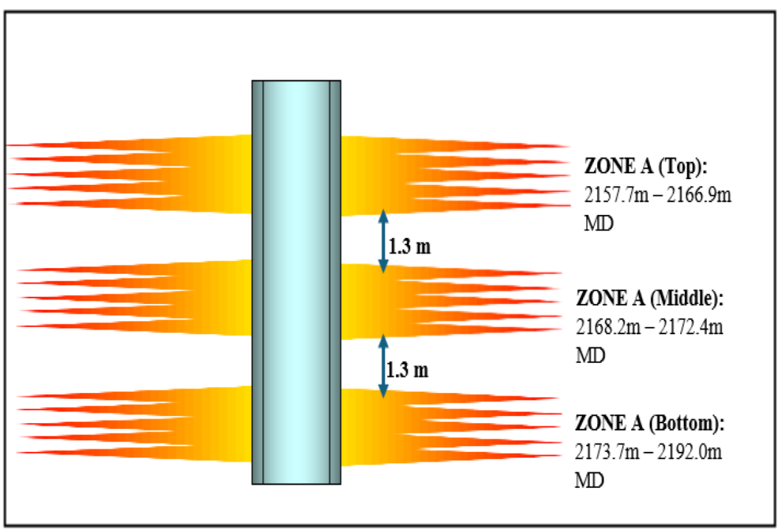



Figure 6—Snapshot of well sketch focusing on Zone A (top/mid/bot).

## Feasibility Study for Sand Control Technique

In view with the direction to ensure proper sand control can be achieved for well K, various technology was evaluated to determine the best sand control technique. The following are the summary of evaluation that was conducted internally for each of the available technology up the decision to proceed with the final technique selection for the well;

Table 2—Sand Control Option Comparison based on internal evaluation.

| Evaluated Technology | Pros | Cons | Cost Vs Economic | Overall Chances of Success |
|---|---|---|---|---|
| Sand Consolidation – Chemical Approach | • Increase rock strength (post treatment).<br>• Provide room for more drawdown post success operation resulted in additional production gain.<br>• Ability to include zone acidizing program as part of sand consolidation program resulted in additional production gain.<br>• No change in completion design as the treatment focusing on formation rather than modifying completion strings. | • Challenges to ensure efficient chemical placement due to close gap between perforation intervals. Proper well preparation required to ensure efficient chemical placement at depth.<br>• To utilized Coiled Tubing package to ensure good chemical placement. | **High** (due to the requirement to have Coiled Tubing with vessel support operation). | **Low** due to,<br>• Level of preparation requirement and dealing with high chemical volume.<br>• No core sample available for lab testing.<br>• Lack of improvement in UCS post acid treatment. |
| Through Tubing Gravel Pack | • Across perforation section sand retention capabilities.<br>• Option available for unwanted sand isolation. | • May require long TTGP stack. Longer TTGP stack may impose issue on integrity and quality.<br>• Hard to have good proppant placement in high deviation segment of the well i.e. 80 deg deviation.<br>• Critical to ensure good conveyance selection to ensure better chances of success.<br>• Various case history show issue in future retrieval for permanent well abandonment. | **High** (due to the requirement to have Coiled Tubing with vessel support operation). | **Low** due to, High well deviation in relation to efficiency in having efficient proppant displacement. |
| Through Tubing Hang-off Screen | • Cheapest option.<br>• Quick and simplified operation as the screen is place above the perforation interval. | • Potential early screen damage due to flow erosion. Increase screen replacement frequency.<br>• Risk of wellbore plugging due to sand accumulation below the screen depth. | **Low** (slickline conveyance to set the screen at low deviation section of the well) | **High** (due to simple operation) |
| High Expansion Screen System | • Screen expansion to tubing wall allow sand retention from wellbore.<br>• Screen compliance to wellbore/tbg ID thus provide better sand retention and minimize plugging<br>• Screen capability for multi grain size sand & fines filtration. | • High well deviation. Precise conveyance equipment is a MUST to ensure success deployment. | **Medium** (Utilizing electric line with tractor and stroker to convey the screen). | **Medium** (no similar operation conducted via eline before) |

After evaluating various options, extensive consideration was given to the chemical sand consolidation approach, supported by detailed laboratory analysis. Due to the unavailability of core samples, laboratory-prepared "sand packs" were constructed based on the mineralogical properties of Sand A. However, out of a total of nine tests conducted, the average unconfined compressive strength (UCS) remained below 1,000 psi, indicating insufficient consolidation, as shown in Table 3.

**Table 3—Chemical Lab Test to confirm Feasibility of Chemical Treatment for Sand A.**

| | #1 | #2 | #3 | #4 | #5 | #6 | #7 | #8 | #9 | #10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Pre-flush / initial perm | 2% KCl | 2% KCl | 2% KCl | 2% KCl | 2% KCl | 2% KCl | 2% KCl | 2% KCl | 2% KCl | 2% KCl |
| Clays | - | 13% | 16% | 17% | 13% | 16% | 17% | 13% | 16% | 17% |
| Surfactant | 0.24% | 0.28% | 0.28% | 0.28% | 0.28% | 0.28% | 0.28% | 0.28% | 0.28% | 0.28% |
| Resin | 24% | 28% | 28% | 28% | 28% | 28% | 28% | 28% | 28% | 28% |
| Curing Agent | 2.80% | 2.80% | 2.80% | 2.80% | 2.80% | 2.80% | 2.80% | 2.80% | 2.80% | 2.80% |
| Core | Grey Berea | 100 mesh | 100 mesh | 100 mesh | 100 mesh | 100 mesh | 100 mesh | 100 mesh | 100 mesh | 100 mesh |
| | | sand pack | sand pack | sand pack | sand pack | sand pack | sand pack | sand pack | sand pack | sand pack |
| Initial perm | 48mD | - | - | - | - | - | - | - | - | - |
| Final perm | 45mD | 13.6D | 4.3D | 15.1D | 6.6D | 6.9D | 15.3D | 7.8D | 14.4D | 17.9D |
| Regained perm | 95% | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A |
| UCS | 1700 | 758 | 692 | 800 | 1126 | 991 | 1145 | 666 | 765 | 748 |
| Original | 3960 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Treated core | 5660 | 758 | 692 | 800 | 1126 | 991 | 1145 | 666 | 765 | 748 |
| Curing Duration | 1 days | 7 days | 7 days | 7 days | 7 days | 7 days | 7 days | 7 days | 7 days | 7 days |

Due to lack of success in chemical sand control/consolidation technique and in line advancement in technology, High Expansion screen system was further evaluated and selected as the option to actively control the sand production due to the following;

a. High expansion screen system capable to hold various particle sizing due to the unique design of the "filtration" unit. Surface Sample Particle Size Distribution (PSD) study was shown in Fig. 7. Based on analysis, it is firmed that this technology are fit for the specific PSD for Sand A.
b. Various extensive experience in conveying electric line and well tractoring for complicated operation such as milling, wellbore cleanout and perforation work build the team confidence level associated to electric line conveyance for high expansion screen installation work.
c. No dependency to vessel for operation execution due to small equipment spread i.e. electric line conveying well tractoring and stroker allow better cost management.

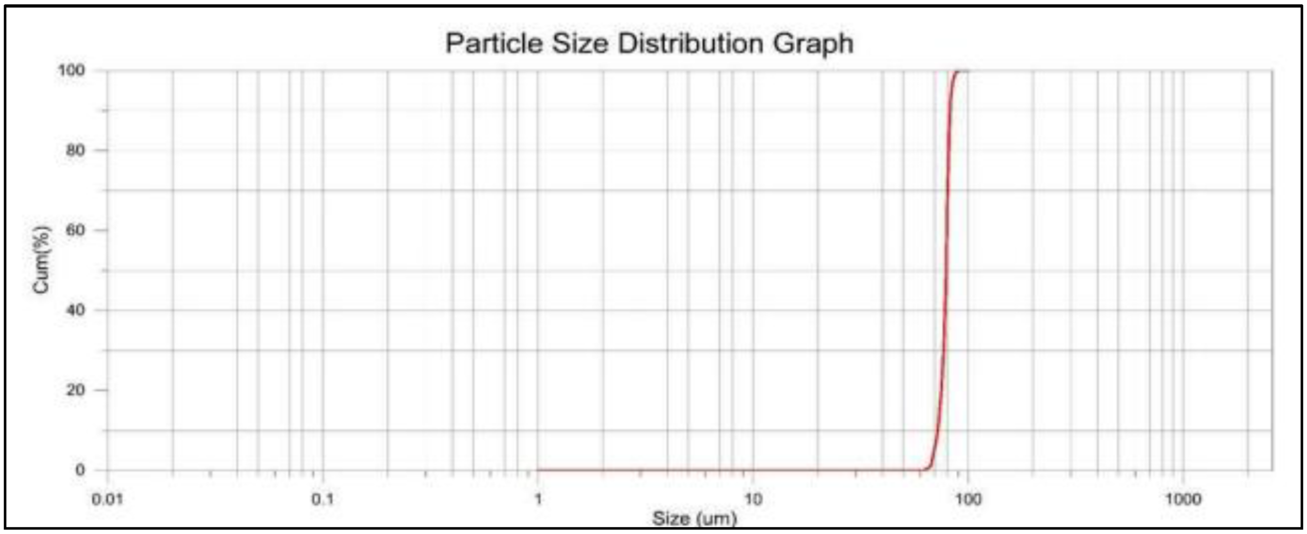



**Figure 7—PSD Information for Sand A. D10 at 114.9µm, D50 at 267.7 µm and D90 at 451.7 µm.**

## High Expansion Screen Technology Introduction

High expansion screen system is a conformable, remedial sand control system which is designed to address the issue of sand failure in the wellbore. Filtration mechanism of the screen is contributed by the expanded "sponge" system which known as multi-layer Open Cell Matrix Polymer (OCMP), which offer various range of sizes to cover broad range of particle and sand sizing.

Fig. 8 show the construction of the OCMP under the microscope. It does not fix to single mesh sizing as such it is matching to Sand A PSD information.

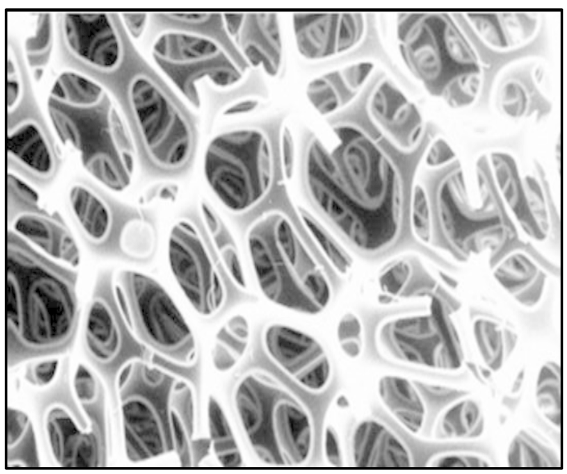


**Figure 8—OCMP construction image under microscope.**

Fig. 9 show the condition of the OCMP when it is fully expanded in the wellbore. The expansion of the OCMP allow sand and particle filtration from entering wellbore from the perforation interval. As and when, the system works properly, it shall be capable to eliminate the risk of having sand and particle deposition in the wellbore which will require further intervention for debris cleanout operation. Total 5m and 2.5m of High Expansion Screen assy./join is available in the market. However, serious consideration shall be made of conveyance prior to selecting the length of the screen as 5 m screen assy. require minimum 4000 lbs of pulling to release the compression sleeve.

Figure 9—OCMP attached to base pipe (uncompressed).

## Running and setting Sequence

As and when the system is deployed in the wellbore, the OCMP will be in compressed mode as it requires to pass through restricted production tubing ID. Various size of the system is available in the market however, for Well K operation 2.69" running OD of high expansion screen was selected to cater for minimum restriction of 2.813" across downhole safety valve.

The unique features of the high expansion screen are it allow single hanging mechanism (profile of non-profile packer) as the suspension point for the screen across even long interval section of the screen. The hanging mechanism to be located at the bottommost section of the screen and all the stack up to be located above the screen.

As shown in Fig. 10, overall setting sequence of the high expansion screen starts with the setting of the bottommost packer which come together with sealbore assemblies, then followed by the run of high expansion screen section. Bottom part of the screen shall come together with the anchor latch assemblies. Maximum allowable set down / jarring down weight limited to 1000 lbs when latching section of the assemblies.

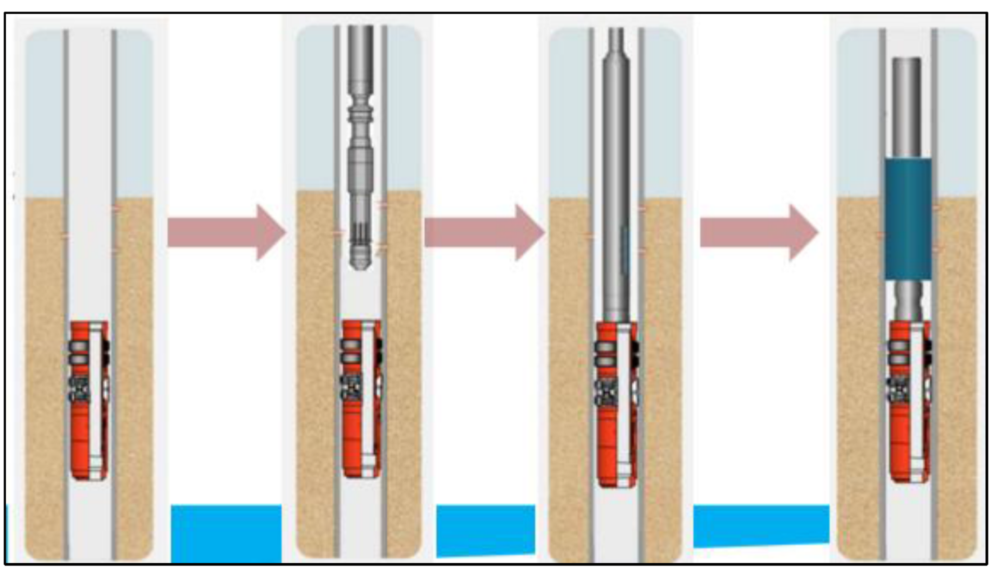


Figure 10—Illustration of High Expansion Screen Setting Sequence

Once the latching process completed, further jarring down / set down weight applied required to shear the pin/screw of the running tools then jarring up required to remove the compression sleeve of the screen. Fully release is established when the compression sleeve is out of the high expansion screen assemblies and the screen is fully expanded to the full tubing wall (as shown in Fig. 9).

Subsequent high screen assemblies stack up to be established from top of the screen that was installed in the well until reaching the highest target depth.

## Well K Job Planning

The job planning for Well K were divided into 2 section which as follows;

a. Minimum high expansion screen length requirement and
b. setting depth.
c. Conveyance planning / Downhole tools selection and configuration.

*Minimum High Expansion Screen Stack Requirement and Setting Depth*

Total length of Zone A is consisting of the following;

Total Zone A perforation length: 31.7 m (to cater for whole Zone A)

Gap between perforation: total 2.6 m

Total Minimum Screen length: 34.3m

In view with the length requirement, total 7 high expansion screen assemblies are required to be set across perforation interval as shown in Fig. 11. Bottom hanging mechanism will be relied on the non-profile packer due to no tubing profile is available across the screen setting depth.

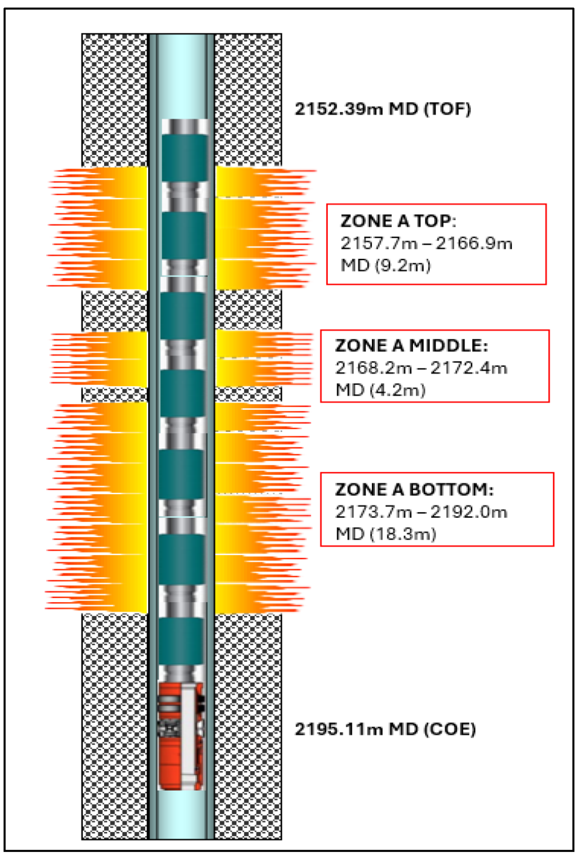



**Figure 11—Overall Setting Assemblies Planned for Well K**

Due to long installation requirement, 5 m of high expansion screen / joint was selected to reduce total no. of runs. Therefore minimum 4000 lbs of pulling capability is required to properly expand the screen in the production tubing.

Based on above consideration, overall stack requirement are as follows;

a. To cater for overall perforation interval length of 31.7m, total 42.72 m of Screen space out to be made up for the well (length inclusive of non-profile plug, latching/sealbore assy and fishing neck.
b. Critical setting parameters for the High Expansion Screen assemblies;
    - Set down weight when latch assembly latching the sealbore to be limited to max. 1000 lbs.
    - Once latching, max. jar down weight to shear the running tools at 2950 lbs +/- 15%.
    - Continue pulling required to retrieve the compressive sleeve out of high expansion OCMP element. Min. 4000 lbs of pulling strength are required to retrieve the compressive sleeve.

**Conveyance Planning / Downhole Tools Selection and Configuration**

As elaborated in previous section, electric line conveyed well tractoring devices is required to set the non-profile packer section and high expansion screen assemblies. However, in view with the need of establishing downhole downward and upward force during setting the High Expansion Screen assemblies, Well Stroker will be required to be included in the bottom hole assemblies.

The selected well stroker for the operation allow maximum of shaft stroke in and out of 20 inch with maximum pulling and pushing capacity of 21,000 lbs which is more than sufficient for the specification of High Expansion Screen.

Knowing specific parameters for High Expansion Screen assemblies setting is crucial to ensure the success of the operation as follows;

a. Maximum setting when latching the assemblies to be limited at 1000 lbs.
b. Once latching, stroker anchor to be set to the tubing wall and stroking out to be continued to approx. 2950 lbs to shear the running tools pin.
c. Inactivate stroker anchor, adjust stroker shaft length to max. length (20"), then stretch eline cable and activate stroker anchor. Stroke in to move the compression sleeve up. Total minimum, 10 cycle is required to move 5 m of compression sleeve out of High Expansion Screen (OCMP Section) body.

After completed running the first assemblies, the proceed is repeated until whole 7 High Expansion Screen assemblies is set in the well.

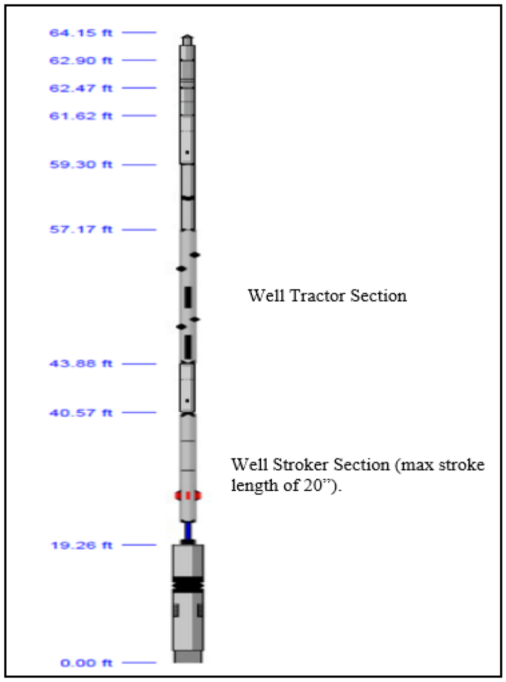



**Figure 12—Downhole tools planning for the operation.**

## 1st Attempt Operation Summaries and Issues

Experience 21% of Non-Productive Time (NPT) during the first job attempt. Not capable to meet the job objective due to the following major issue;

### Issue#1: Post successful setting High Expansion Screen#1, having issue to pass through TRSCSSV when Pull Out to to Surface

During the operation, set down weight at well tractor was set to maximum setdown weight of 1000 lbs. Continue, stroke out to shear the running tools pin was done without any issues.

Overall stroker activation to release compression sleeve was executed as per plan it it shown clearly in the chart (as shown in Fig. 13). Total 12 stroker cycle was executed to let the running tools free which indicate the compression sleeve is out of the high expansion screen section assemblies. When pull out of hole, the tools is stuck across TRSCSSV, push the decision to activate the stroker and pull to maximum of 7,000 lbs (downhole) to get the whole assemblies out of wellbore.

**Figure 13—— Stroker activation cycle to set High Expansion Screen**

As and when the tools arrived surface, visual inspection conducted, and it was observed the bottom part of compression sleeve are expanded to 2.82" OD with expanded length of 3" from bottom of the compression sleeve (as shown in Fig. 14).

**Figure 14—– Expanded bottom part of compression sleeve.**

**Issue#2: Having issue to have a good latch during High Expansion Screen#2 run**

After the recovery of the stuck compression sleeve, 2nd run was made to convey 2nd High Expansion Screen section to the depth. When arriving to the top of screen section, multiple attempts were made in order to latch the sealbore assemblies of screen section. In view of the issue, it was decision to pull out of well and recheck the tools condition. Then, rerun was conducted to continue with latching attempt. Total 3 runs were conducted for the same purpose without any success.

In view with the failure, Lead Impression Block (LIB) was also conveyed in the well to confirm in case of centralization play role in the issue in latching the upper section of the screen to the bottom section of the screen which located in the wellbore.

Fig. 15 show the condition of the latch assemblies when it reach surface post multiple latching attempts. There was no marking on the latch mechanism to conclude potential issue with tool centralization issue.

**Figure 15—Latch mechanism for screen.**

In view with the issue above, it was decided to temporarily stop the operation and conducted further investigation. All equipment was send back to base for investigation.

## Investigation Finding, Conclusion and Way forward

Thorough Based investigation was conducted post operational failure. The investigation was mainly focused to re-simulate the failure that was experience during the operation involving issue during the operation. Then, further improvement based on actual finding and the root cause.

### Base Inspection and Finding

Failure simulation for Issue#1 involve of the process to make and attempt to push the compression sleeve towards bottom section until the end ring assemblies for the high expansion sleeve and identify total force required to move the compressive sleeve towards the end ring assemblies.

While for Issue#2, there are 2 separate test conducted at separate location due to the supplier of the services and item come from different organization which as follows;

a. High Expansion Screen sealbore and latch assemblies latch test conducted to all assemblies it is for the purpose of latch-ability check.
b. Testing the well stroker functionality Vs its capability to deliver required force.

Based on base/lab test, the following finding was observed and direct conclusion made from the finding.

| Issue | Lab Simulation Finding | Conclusion |
|---|---|---|
| #1 | Compression sleeve movement to overlap the high expansion sleeve end ring when the 6500 lbs of set down weight applied (as show in Fig. 16) | There are chances of even the limit was set @ 1000 lbs (on wireline), spike in inertia and momentum (due to tools string weight) when tools are touching the seal contribute to excessive weight. |
| #2 | Surface test confirm that the latching assy. not be able to latch to the sealbore assemblies. | Potential error with equipment (latch/sealbore) design/ manufacturing. |
| #3 | Surface test of stroke out and stroke to 3K lbs. | No issue. Weight load indicator confirms load transfer as per plan. |

Further investigation on Issue#2 lead to the following findings;

a. There was major discrepancy in latch assy/sealbore design Vs Manufacturing & QAQC process prior to equipment delivery to client. There are mistake in drawing revision between latch assy vs sealbore section. During manufacturing, outdate engineering drawing was input into the CNC machine which causing discrepancy in design between sealbore and latch assemblies' section.
b. There was no mechanism to check the latch-ability as part of QAQC process post manufacturing. Once the tools latching, it will cause the compression sleeve required to be pulled out of the high Expansion screen section then the OCMP will be fully expanded (unusable).
c. Long High Expansion Screen stack was the first experience even by the Services provider. Therefore, in Operator conclusion Service Provider did not take serious consideration on latching assy/seal bore test as part of QAQC plan.

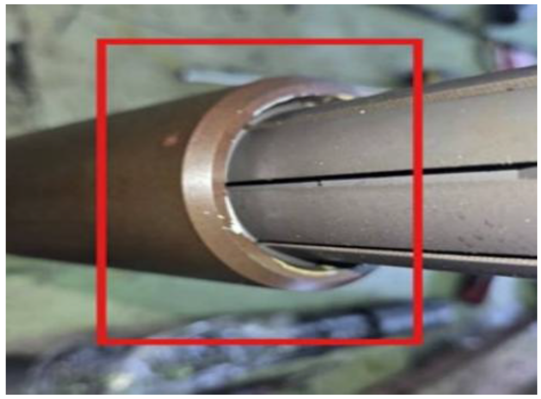


Figure 16—Latch assy. to seal bore section at screen.

### Improvement Plan and Operational Way Forward

Few main improvement plans were debated and agreed prior to the deployment of the corrected High Expansion Screen Assemblies as follows;

| Issue | Improvement Plan |
|---|---|
| #1 | a. Maintaining Eline wire pull to 1000 lbs while activating tractor in order to eliminate the momentum while latching the tools.<br>b. Latching assy. to be in contact with sealbore via tractor force ONLY during latching process (approx. 500 lbs of tractoring force). |
| #2 | a. Return all remaining High Expansion Screen Assy. to service provider for re-manufacture.<br>b. Services providers establish latch-ability check procedure prior to equipment delivery & prior to sending offshore.<br>c. Review all QAQC plan to ensure proper item delivered and re-do all the surface testing prior to running the item into wellbore. |

## 2nd Attempt Operation Summaries

All mitigation measures implemented during the operation proved effective, ensuring smooth execution and successful achievement of the job objectives as planned. The second attempt was completed with zero Non-Productive Time (NPT), reflecting the strength of the revised strategy.

Total expenditure was approximately 8% higher than the approved Authorization for Expenditure (AFE) due to the need to remobilize electric line and well tractor services for the second attempt.

No sand production was observed during the well's return to production. The well was successfully brought back to its original tubing pressure (TP) with a stable gas rate of 6 million standard cubic feet per day (mmscfd).

This operation marked a global first: the successful setting of a long High Expansion Screen via electric line conveyed by a Well Tractor and Stroker in a highly deviated well.

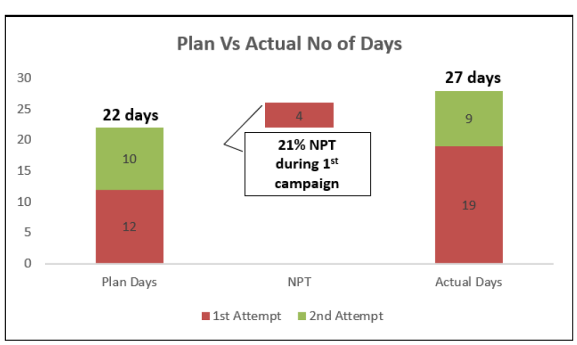



Figure 17—Plan Vs Actual Operational Days.

Numerous valuable lessons were gained throughout the process, which can be shared with other operators facing similar downhole challenges.

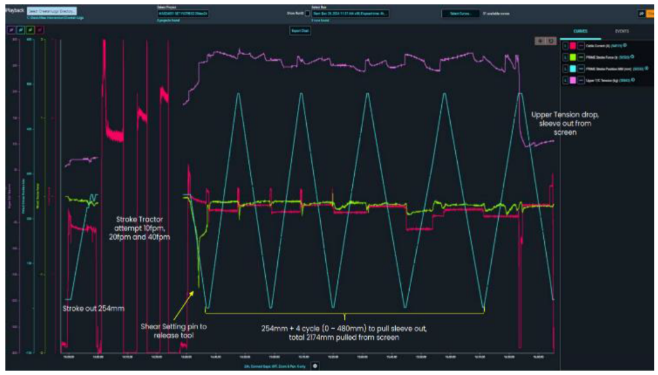



**Figure 18—Improvement sequence of screen setting.**

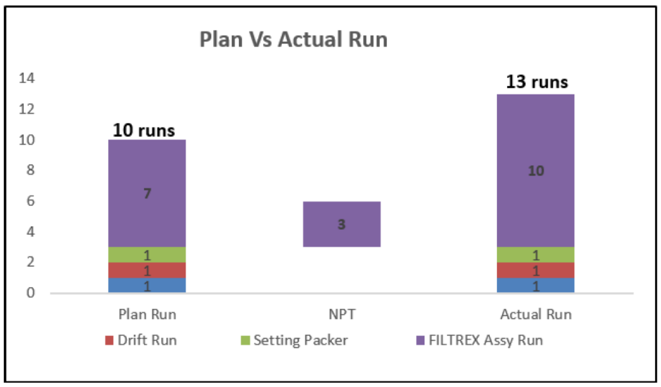



**Figure 19—Plan Vs Actual No. of runs.**

## Conclusions

Despite of having challenges in operation due to unable to identify the potential risk upfront during the job plan, Well K High Expansion Screen Conveyance was successfully executed:

- There was no HSE related incident encountered during the operation. All the crew involved in the operation work hand in hand to ensure proper job implementation in a safe manner.
- Various lesson learnt was gathered during the operation which can become future reference during similar job application.
- The was success in production delivery back to the well original technical potential.
- Close monitoring equipment design Vs equipment manufacturing led to right path in product development.
- In-house development program proven to deliver success to the operation. Subsequent 2 similar operation was conducted recently with success and no issue with tools stuck in hole.

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